

EXAMPLE 3.2

Outcomes

- Binary property analysis
- Model versus TDE data
- RadFrac distillation modelling
- Column profiles
- Plot graphs

Problem Description

A distillation column is used to separate water and methanol. The feed flowrate has the following parameters: -

63.2 wt.% water

36.8 wt.% methanol

Total flow: 120,000 lb/hr

Pressure: 18 psia

Saturated liquid

The column specifications are as follows: -

38 trays (40 stages)

Feed tray: 23 (stage 24)

Total condenser

Top stage P:16.1 psia

Distillate flowrate: 1245 lbmol/hr

Molar reflux ratio: 0.5

Using NRTL-RK, model the process using RADFRAC model.

1. Start a new simulation using the Blank Simulation template in Aspen Plus.
2. In the Components | Specifications | Selection sheet enter METHANOL in the Component ID column. Note that Component name and Alias are automatically filled for this component. Enter WATER as Component ID for the second component.

Component ID	Type	Component name	Alias	CAS number
METHANOL	Conventional	METHANOL	CH4O	67-56-1
WATER	Conventional	WATER	H2O	7732-18-5

Find Elec Wizard SFE Assistant User Defined Reorder Review

- Next define property methods. Click on Methods in the navigation pane. In the Method filter option select ALL to list all thermodynamic method in the Base method and select NRTL-RK. The NRTL-RK uses NRTL method for liquid phase and RK method for gas phase. It is typically used to model systems containing polar components at low to medium pressure.

- Click NEXT to retrieve the binary interaction parameters from Aspen database. You do not need to enter any other values.

Binary Interaction - NRTL-1 (T-DEPENDENT) X +

Input

Databanks

Comments

Parameter NRTL Help Data set 1 Swap Enter Dechema Format Estimate using UNIFAC View Regression Information Search BIP Completeness

Temperature-dependent binary parameters

Component i	Component j	Source	Temp. Units	A11	A12	A13	A21	A22	A23	A31	A32	A33	A41	A42	A43	A44	A45	A46	A47	A48	A49	A50	A51	A52	A53	A54	A55	A56	A57	A58	A59	A60	TLOWER	TUPPER
METHANOL	WATER	APV121 VLE-RK	C	-2.626	4.8241	828.387	-1329.54	0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	24.99	188.3	

- Property analysis. In the Analysis section of the Home ribbon, select Binary to open the Binary Analysis window. Select Analysis type, Txy. In the Components section, the lighter component should be entered in the component list first. Therefore, Component 1 is methanol and Component 2 is water.

Binary Analysis | Tabulate | Calculation Options | Diagnostics | Results | Comments | Status

Analysis type: **Txy**

Components

Component 1: **METHANOL**

Component 2: **WATER**

Pseudo-Binary

☐ Pseudo-binary system

Entrainer:

Entrainer fraction:

Compositions

Basis: **Mole fraction**

Vary: **METHANOL**

☒ Equidistant ☐ Logarithmic ☐ List of values

Start point:

End point:

☒ Number of intervals:

☐ Increment:

Pressure

Units: **bar**

☐ Equidistant ☒ List of values

Enter Values:

Run Analysis

- The Calculation Options tab opens the Flash options and Property options sections. Use this to change the valid phases, property method and other settings if necessary.

Binary Analysis | Tabulate | Calculation Options | Diagnostics | Results | Comments | Status

Flash options

Valid phases: **Vapor-Liquid-Liquid**

Maximum iterations:

Error tolerance:

Flash convergence algorithm:

☒ Use flash retention

Property options

Property method: **NRTL-RK**

Henry components:

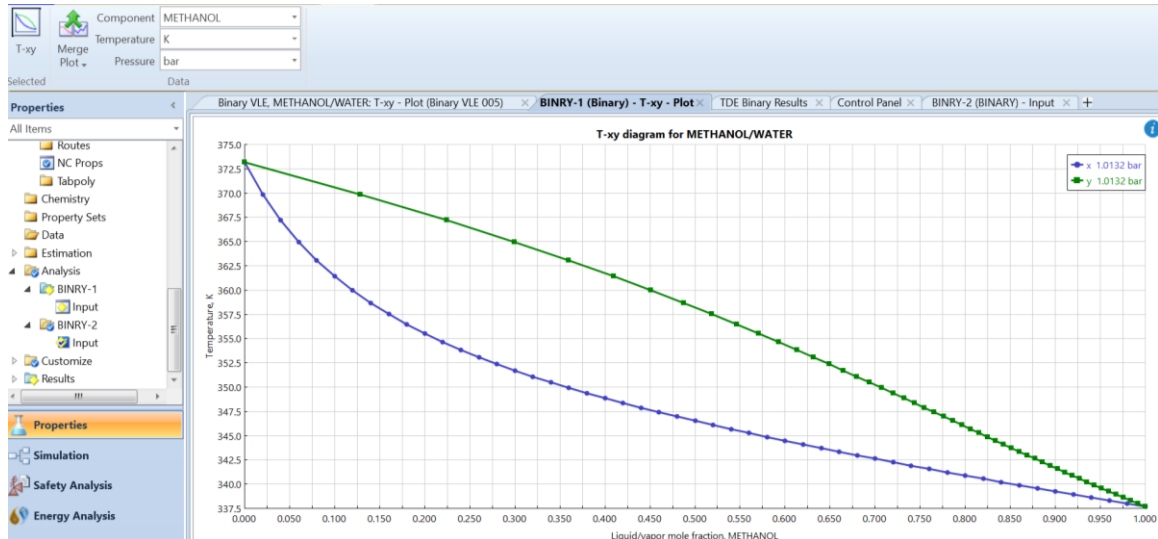
Chemistry ID:

Calculation approach: **True components**

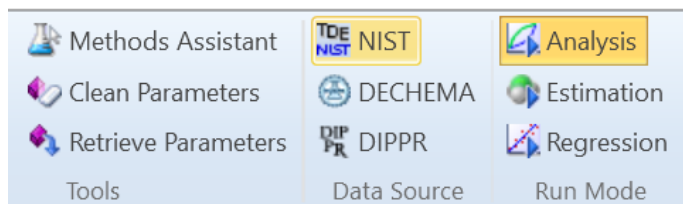
- Return to the Binary Analysis tab and click Run Analysis button. On the Design ribbon change the Temperature to K. The created T-xy plot is shown below.

INDUSTRIAL PROCESS MODELLING & SIMULATION TRAINING

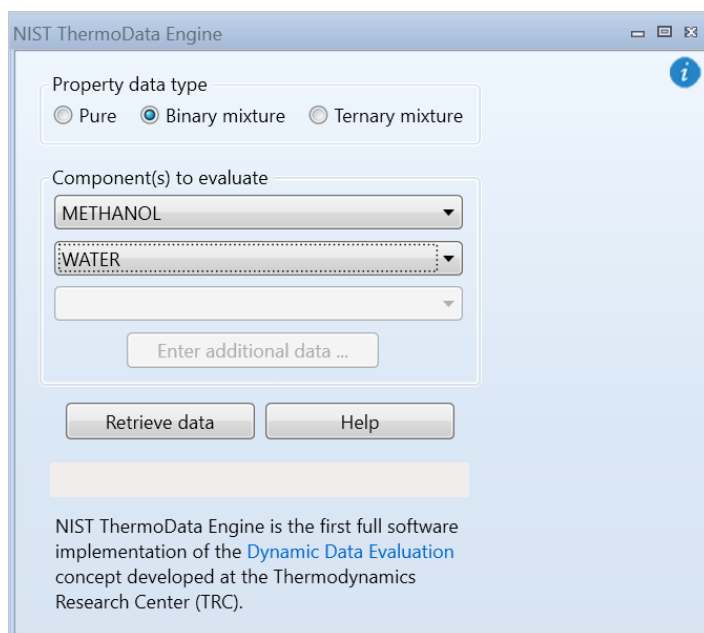
PART 3: DISTILLATION MODELLING & SIMULATION



8. To compare the model with experimental data, at the Home ribbon select NIST to open the ThermoData Engine (TDE). NIST-TDE is a thermodynamic data correlation, evaluation, and prediction tool provided with Aspen Plus and Aspen Properties. The purpose of the TDE software is to provide critically evaluated thermodynamic and transport property data based on the principles of dynamic data evaluation



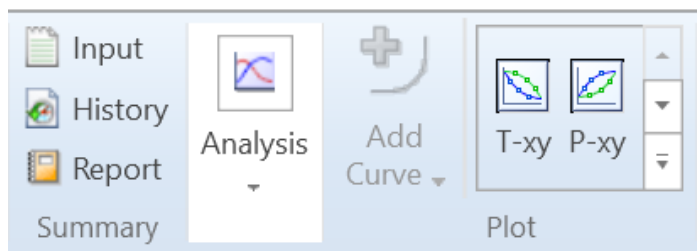
9. On the NIST ThermoData Engine window, select Binary mixture for the Property data type. Select methanol and water as components to evaluate. Click Retrieve data button.



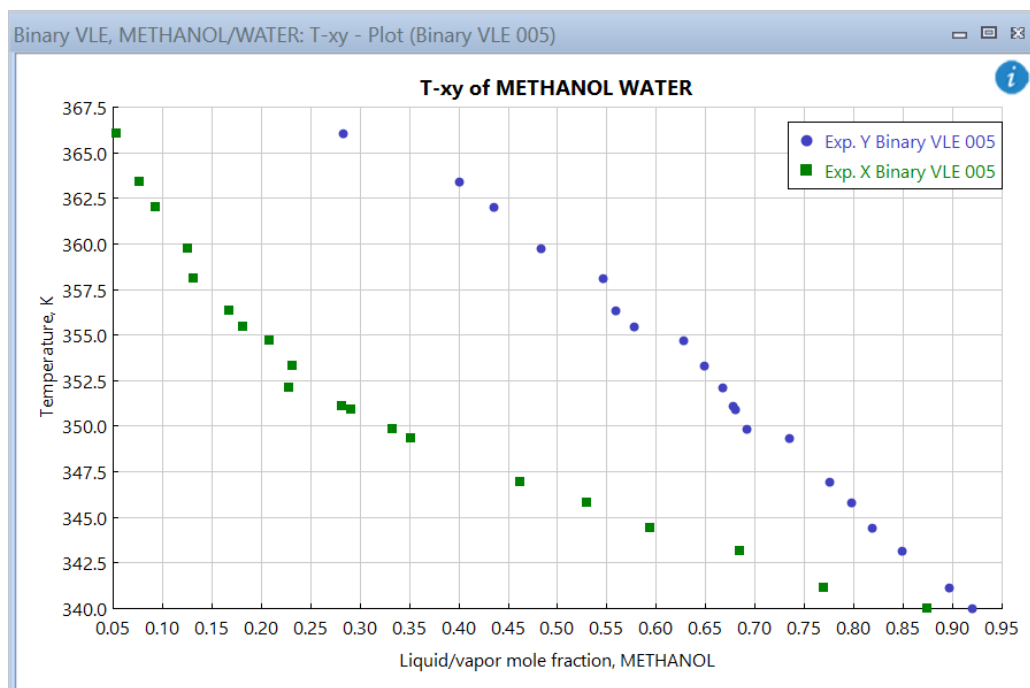
10. On the TDE Binary Results window on the Experimental Data sheet go to Binary VLE | Isobaric select Binary VLE 005 to view the experimental data.

No.	Liquid mole fraction METHANOL	Temperature (K)	Vapor mole fraction METHANOL	Total pressure (N/sqm)
1	0.0531	366.0233	0.2834	101320
2	0.0767	363.423	0.4001	101320
3	0.0926	362.023	0.4353	101320
4	0.1257	359.723	0.4831	101320
5	0.1315	358.123	0.5455	101320
6	0.1674	356.323	0.5585	101320
7	0.1818	355.423	0.5775	101320
8	0.2083	354.723	0.6273	101320
9	0.2276	352.1231	0.6674	101320
10	0.2319	353.323	0.6485	101320
11	0.2818	351.1232	0.6775	101320
12	0.2909	350.9232	0.6801	101320
13	0.3333	349.8233	0.6918	101320
14	0.3513	349.3234	0.7347	101320
15	0.462	346.9236	0.7756	101320

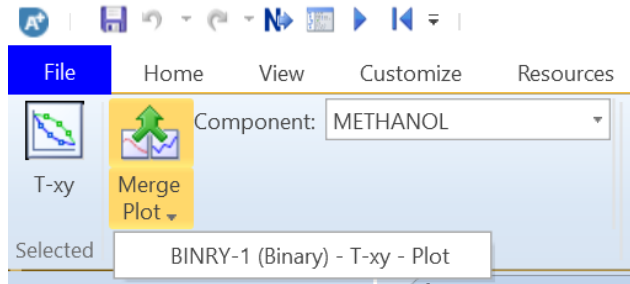
11. To plot the Txy diagram at the Home ribbon go to the Plot tool and select T-xy and the plot pops up.



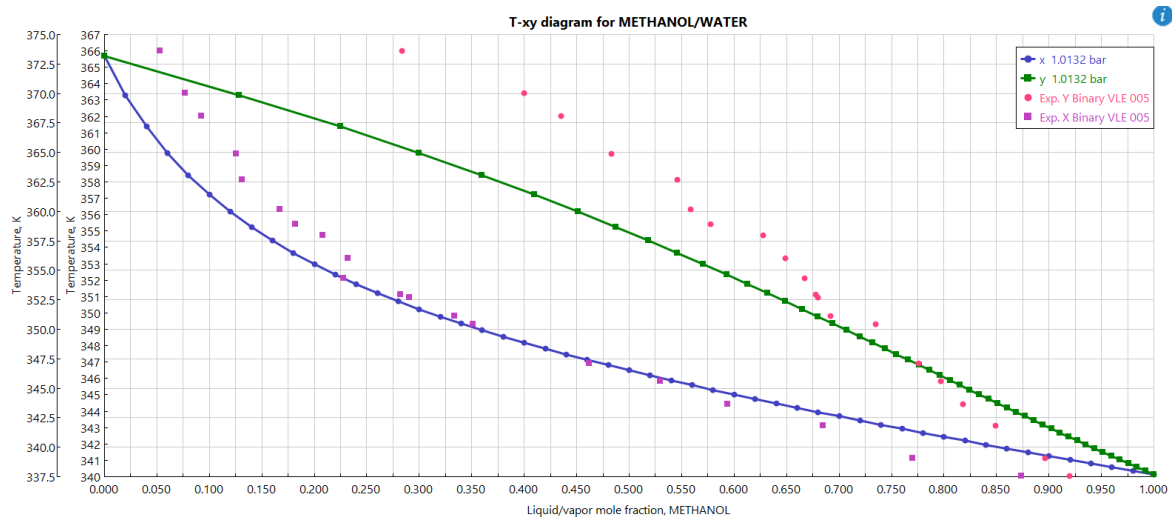
12. Right click on the title bar and select Tabbed to arrange the plot as Tab in the main window.



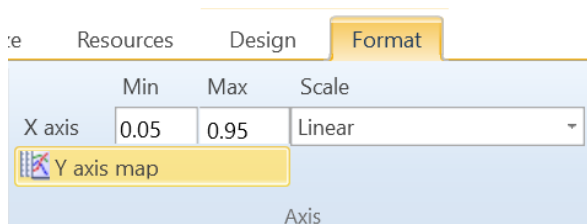
13. To merge the TDE plot with NRTL-RK model, on the Plot | Design ribbon click Merge Plot and select the NRTL-RK Binary plot.



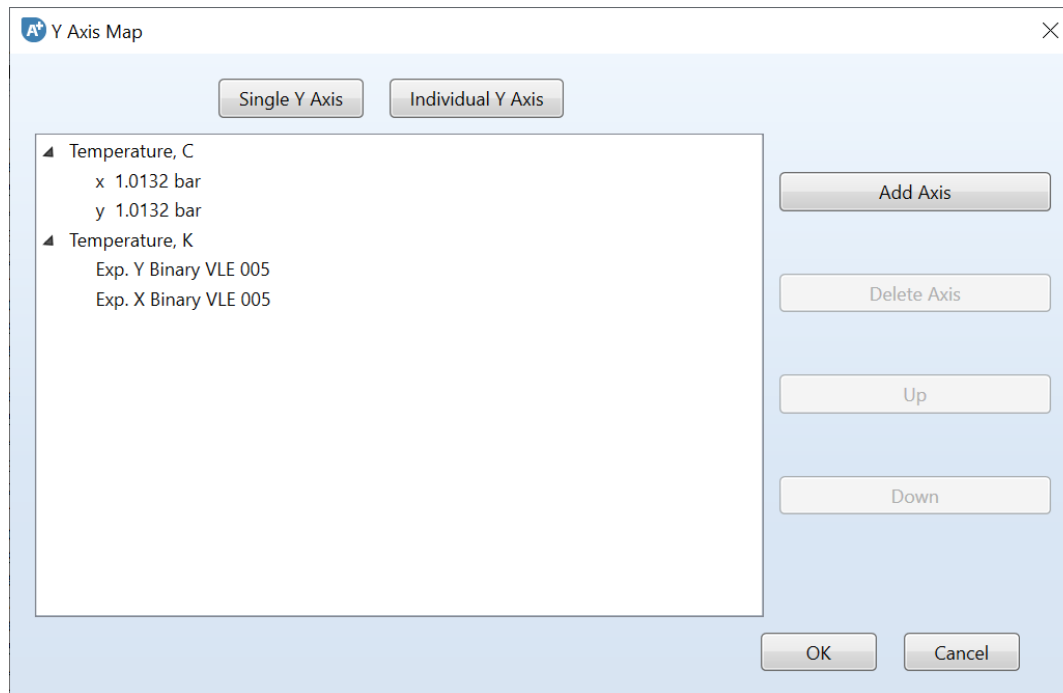
The merged plot is shown below.



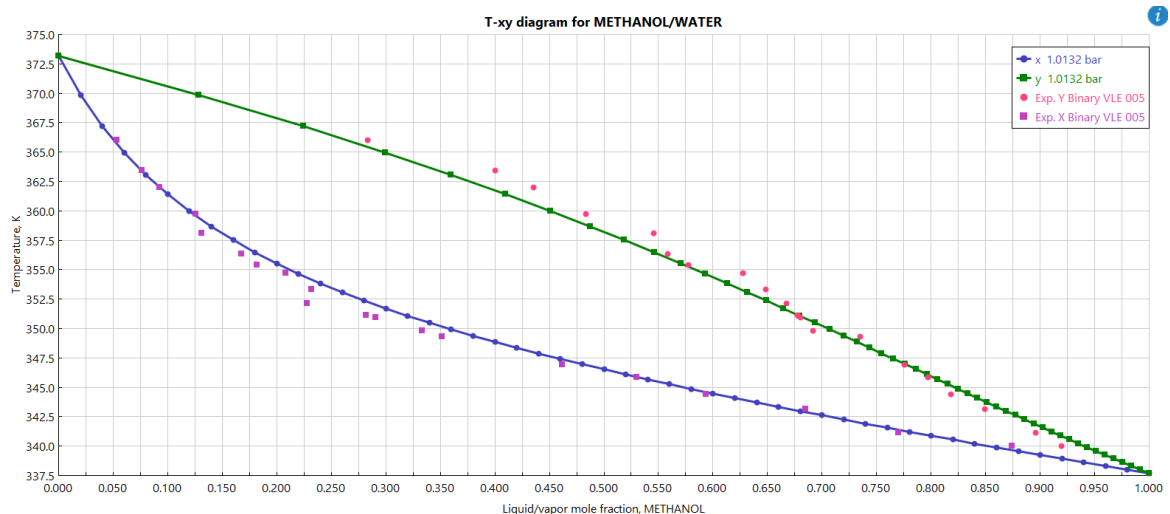
14. The X-axis for temperature needs to be combined for better visualization. On the Plot | Format ribbon select Y axis map.



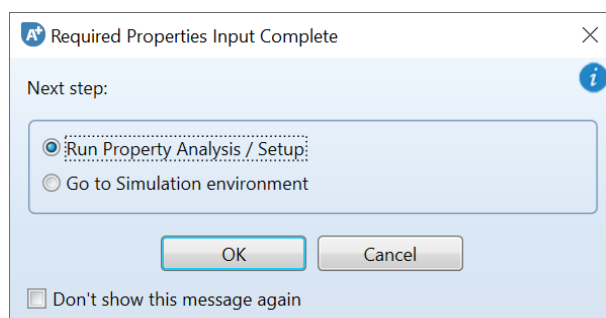
15. On the Y Axis Map window select Single Y axis and click OK.



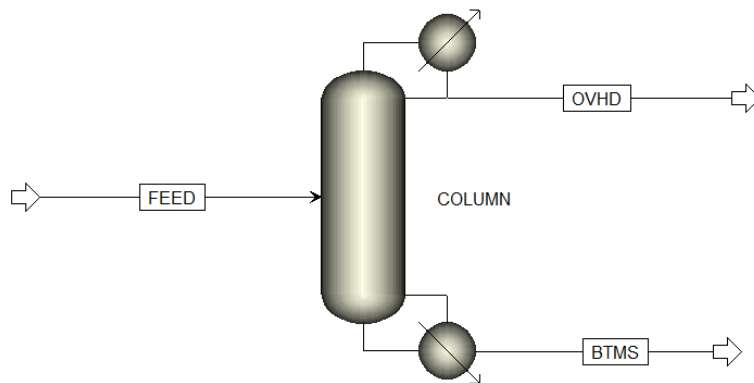
The new plot is shown below. The model fitted well with the NIST experimental data and therefore NRTL-RK is a good method for the simulation.



16. Click NEXT to Run the Property Analysis / Setup. Click the Control Panel option to view the property run results.



17. Go to the simulation environment by clicking on the Simulation button below the navigation pane.
18. To create the process flowsheet as shown below, add the RADFRAC model located under the Columns tab in the Model Palette, and name it COLUMN. Connect the inlet and outlet ports with Material streams and name them accordingly. Note for total condenser, only a single stream is needed to be connected to the liquid distillate port.



19. Double-click the FEED stream to enter the input stream characteristics. Select Vapor Fraction and Pressure for Flash Type. Enter 18 psia for Pressure and 0 for Vapor fraction. Change the Total flow basis to Mass and the units to lb/day. Enter 120,000 into the Total flow rate field. Change the Composition basis to Mass-Frac and enter 0.368 for methanol and 0.632 for water.

Main Flowsheet x FEED (MATERIAL) x +

Mixed CI Solid NC Solid Flash Options EO Options Costing Comments

Specifications

Flash Type **Vapor Fraction** **Pressure**

State variables

Temperature C

Pressure **psia**

Vapor fraction

Total flow basis **Mass**

Total flow rate **lb/hr**

Solvent

Reference Temperature

Volume flow reference temperature C

Component concentration reference temperature C

Composition **Mass-Frac**

Component	Value
METHANOL	0.368
WATER	0.632

Total

20. Double-click on the column to enter the specifications. In the Configuration sheet, select Equilibrium for the Calculation type. Enter 40 into the Number of stages field. The numbering convention in Aspen Plus is that the condenser is stage 1, with stage numbers increasing down the column. For Condenser type select Total. Remain the selection for Reboiler, Valid phases and Convergence fields. For Operating specifications, two variables need to be specified. Initial estimations will be used as first approximations, which can then be refined in order to meet product specifications. Select Distillate rate (mole in lbmol/hr) and Reflux ratio (mole) and enter 1245 and 0.5 respectively.

COLUMN (RadFrac) x COLUMN (RadFrac) - Stream Results (Boundary) x Main Flowsheet x FEED (MATERIA

☒ Configuration ☐ Streams ☐ Pressure ☒ Condenser ☒ Reboiler ☐ 3-Phase ☐ Comments

Setup options

Calculation type: *Equilibrium*

Number of stages: **40** [Stage Wizard]

Condenser: **Total**

Reboiler: *Kettle*

Valid phases: *Vapor-Liquid*

Convergence: *Standard*

Operating specifications

Distillate rate: **Mole** **1245** **lbmol/hr**

Reflux ratio: **Mole** **0.5**

Free water reflux ratio: **0** [Feed Basis]

[Design and specify column internals]

21. Next go to the Streams sheet. In the feed stage field enter 24. There are four types of convection. Above-Stage is selected for vapor from feed goes to stage above feed stage and liquid goes to feed stage. On-Stage, Liquid and Vapor is selected for vapor and/or liquid from feed go to specified feed stage. For most realistic column scenarios, Above-Stage should be used.

☒ Configuration ☒ Streams ☐ Pressure ☒ Condenser ☒ Reboiler ☐ 3-Phase ☐ Comments

Feed streams

Name	Stage	Convention
FEED	24	<i>Above-Stage</i>

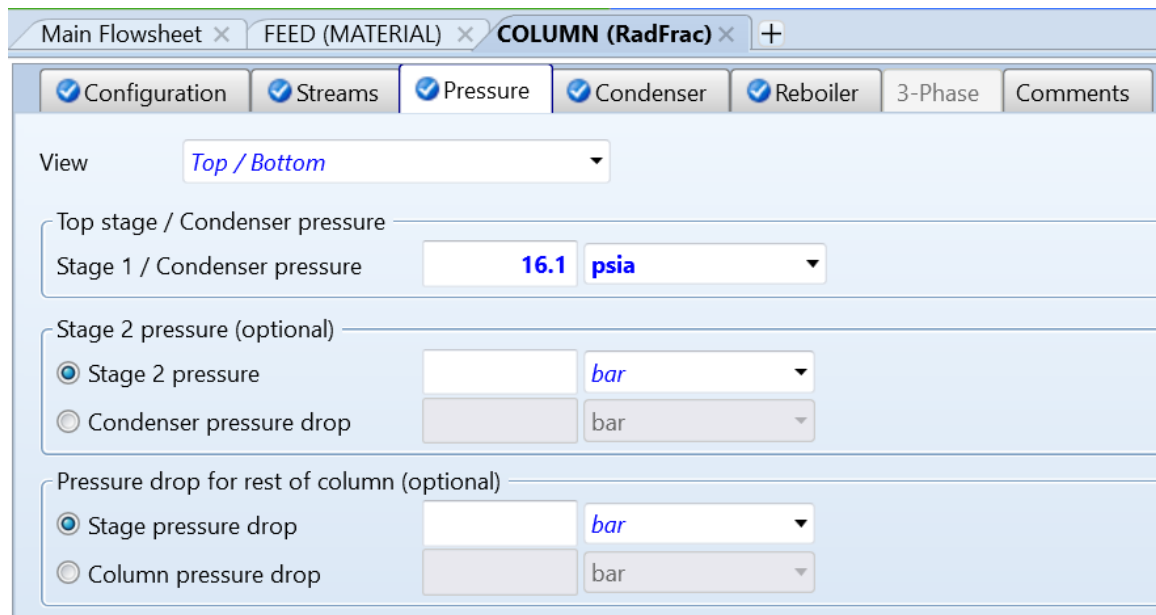
Product streams

Name	Stage	Phase	Basis	Flow	Units	Flow Ratio	Feed Specs
BTMS	40	Liquid	Mole		kmol/hr		Feed basis
OVHD	1	Liquid	Mole		kmol/hr		Feed basis

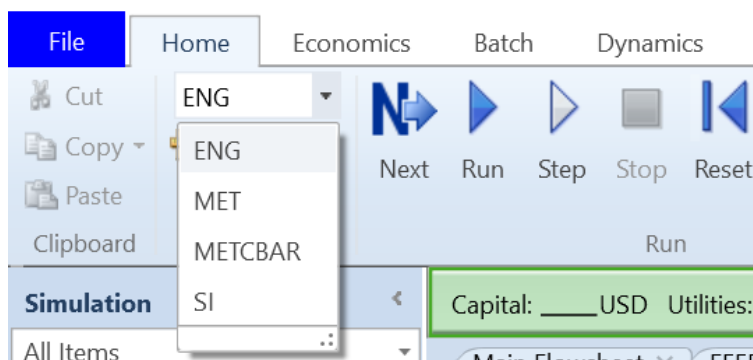
Pseudo streams

Name	Pseudo Stream Type	Stage	Internal Phase	Reboiler Phase	Reboiler Conditions	Pumparound ID	Pumparound Conditions	Flow	Units
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22. Go to Pressure sheet. Enter 16.1 in the Stage 1 / Condenser pressure and change the unit to psia. For this problem, we will assume that there are no pressure gradients throughout the column. If no pressure drop is entered into Aspen Plus, the top stage pressure will be assumed to be the operating pressure of the entire column. Note that in real life situations there will be a pressure drop throughout the column.



23. Open the Control Panel and run the simulation by pressing the run button (F5). To view the column results, Go to the Blocks | COLUMN | Results | Summary sheet. To change the unit, go to Home ribbon. On the Global Units section, select the desired unit sets. For this example, use ENG unit.



24. To check stream results. Go to the Blocks | COL | Stream Results | Material sheet.

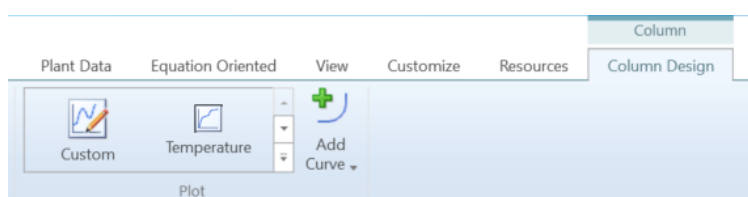
COLUMN (RadFrac) x		COLUMN (RadFrac) - Stream Results (Boundary) x				Main Flowsheet x		FEED (MATERIAL) x		Control	
Material	Heat	Load	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids				
					Units	FEED	BTMS	OVHD			
	Description										
	From						COLUMN	COLUMN			
	To					COLUMN					
	Stream Class					CONVEN	CONVEN	CONVEN			
	Maximum Relative Error										
	Cost Flow				\$/hr						
	- MIXED Substream										
	Phase					Liquid Phase	Liquid Phase	Liquid Phase			
	Temperature				F	186.773	196.273	158.8			
	Pressure				psia	18	16.1	16.1			
	Molar Vapor Fraction					0	0	0			
	Molar Liquid Fraction					1	1	1			
	Molar Solid Fraction					0	0	0			
	Mass Vapor Fraction					0	0	0			

25. To view the column profile to Blocks | C2SPLIT | Profiles. Here you can view Temperature, Pressure, Heat, and Material profiles over the column. Clicking the Compositions tab will display the compositions at each stage in the column. You can choose to display either vapor or liquid compositions, and you can choose to display mole or mass fractions.

The screenshot shows the Aspen Plus software interface. On the left, the 'All Items' tree is visible with 'COLUMN' expanded, and 'Profiles' selected. The main window is titled 'COLUMN (RadFrac) - Profiles'. It has tabs for 'TPFQ', 'Compositions', 'K-Values', 'Hydraulics', and 'React'. The 'View' dropdown is set to 'Liquid' and the 'Basis' dropdown is set to 'Mass'. A table displays the liquid composition profiles for six stages.

Stage	METHANOL	WATER
1	0.863385	0.136615
2	0.633999	0.366001
3	0.474437	0.525563
4	0.402343	0.597657
5	0.376685	0.623315
6	0.368321	0.631679

26. You can then graph these profiles by using the plot tool located on the Column Design ribbon.



27. To plot column composition in the Plot tool, drop down menu select Compositions. A Compositions profiles window pops out. Select phase as Liquid and Mole as basis. Select methanol and water as components. Click OK to view the compositions plot.

